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Yang et al.

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(54) **DIGITAL-TO-ANALOG CONVERSION CIRCUIT AND METHOD**

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(57) **ABSTRACT**

Related U.S. Application Data

(63) Continuation of application No. PCT/CN2021/079754, filed on Mar. 9, 2021.

Provided are a digital-to-analog conversion circuit and method. The digital-to-analog conversion circuit includes a conversion unit and a first regulating unit. The conversion unit includes a first code value receiving terminal used to receive a first code value, and the conversion unit is used to convert the first code value into an analog signal. The first regulating unit includes a second code value receiving terminal used to receive a second code value. A signal input terminal of the first regulating unit is connected to a signal output terminal of the conversion unit. The first regulating unit is used to acquire a first analog regulation signal according to the second code value, and regulate, by using the first analog regulation signal, an analog signal transmitted by the conversion unit.

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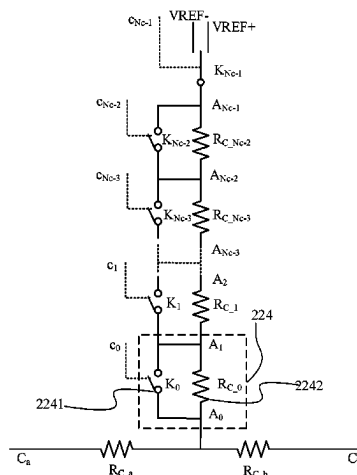
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H03M 1/78 (2006.01)
H03M 1/06 (2006.01)

(52) **U.S. Cl.**
CPC **H03M 1/0604** (2013.01)

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H03M 1/808; H03M 1/0682;

(Continued)

16 Claims, 10 Drawing Sheets



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H03M 1/069; H03M 1/12; H03M 1/66;
H03M 1/664; H03M 1/70; H03M 1/747

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See application file for complete search history.

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100

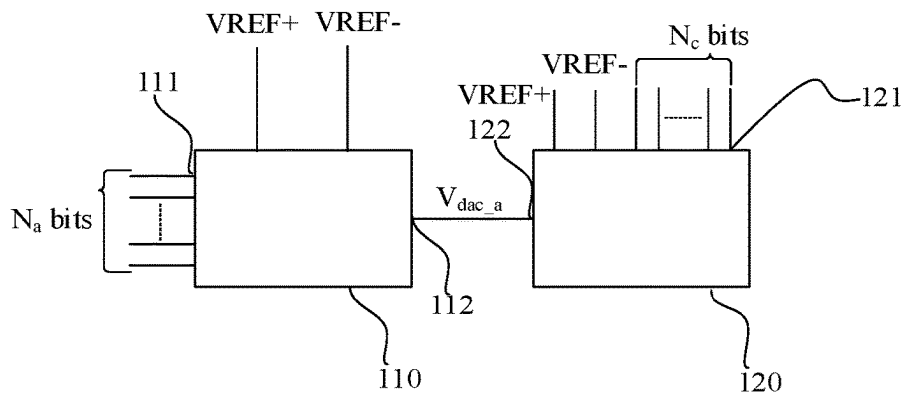


FIG. 1

100

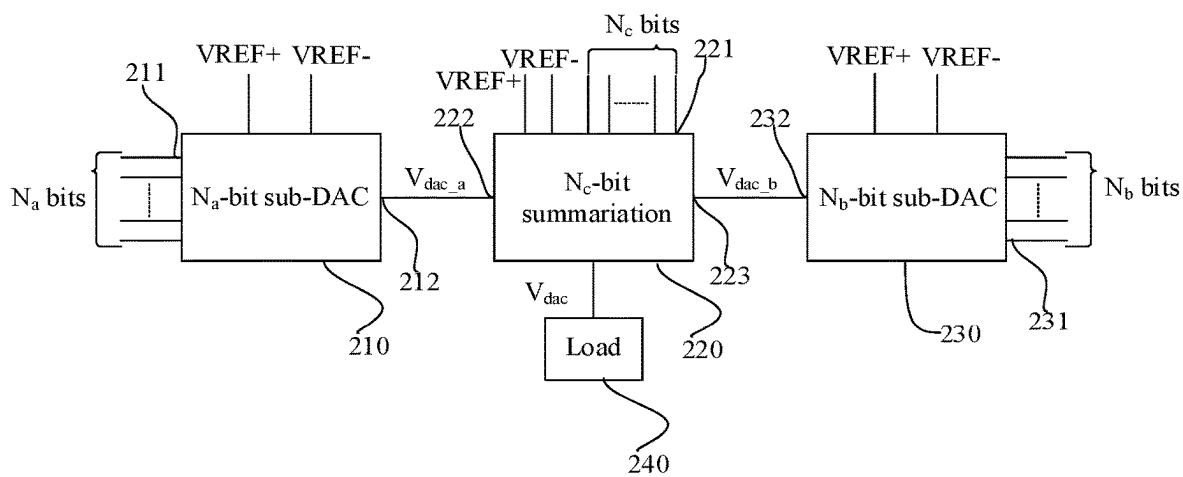


FIG. 2

240

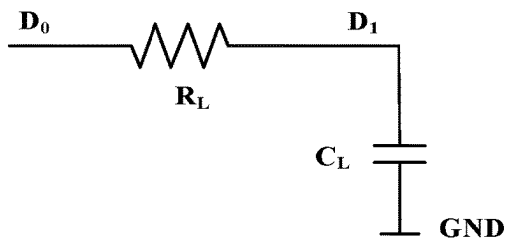


FIG. 3

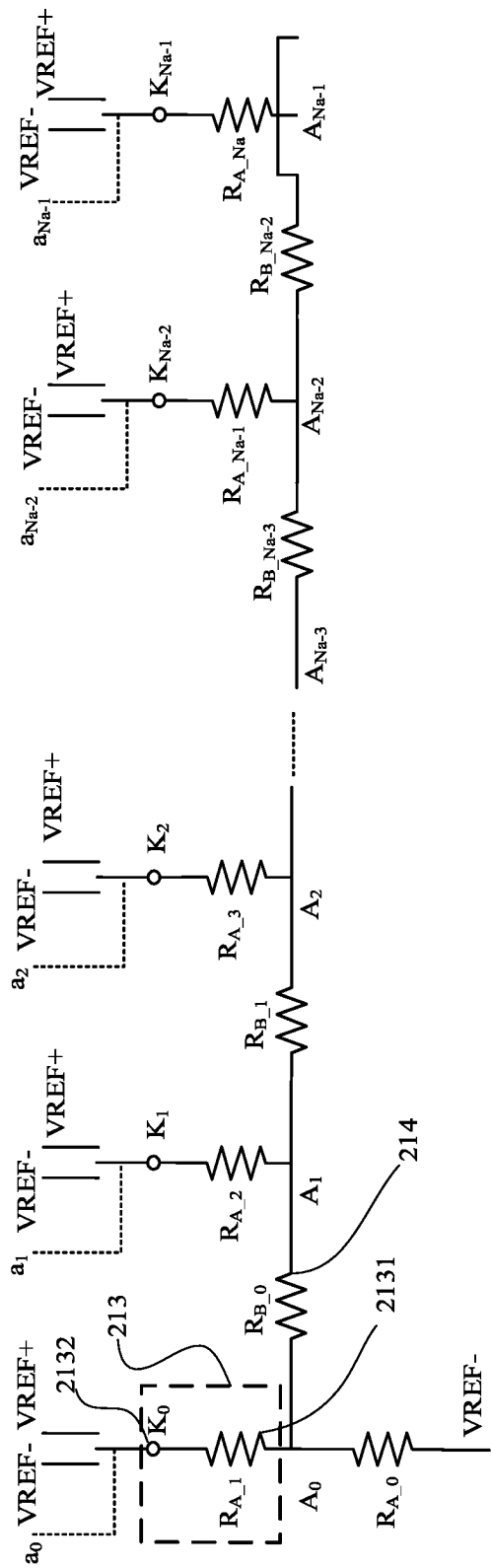


FIG. 4

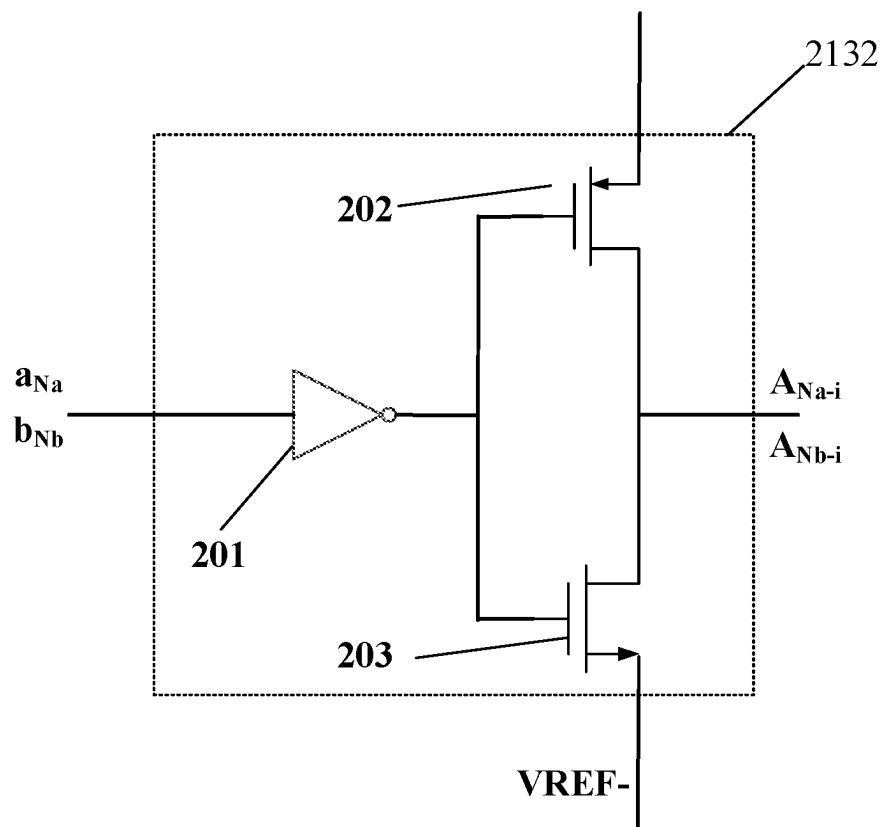


FIG. 5

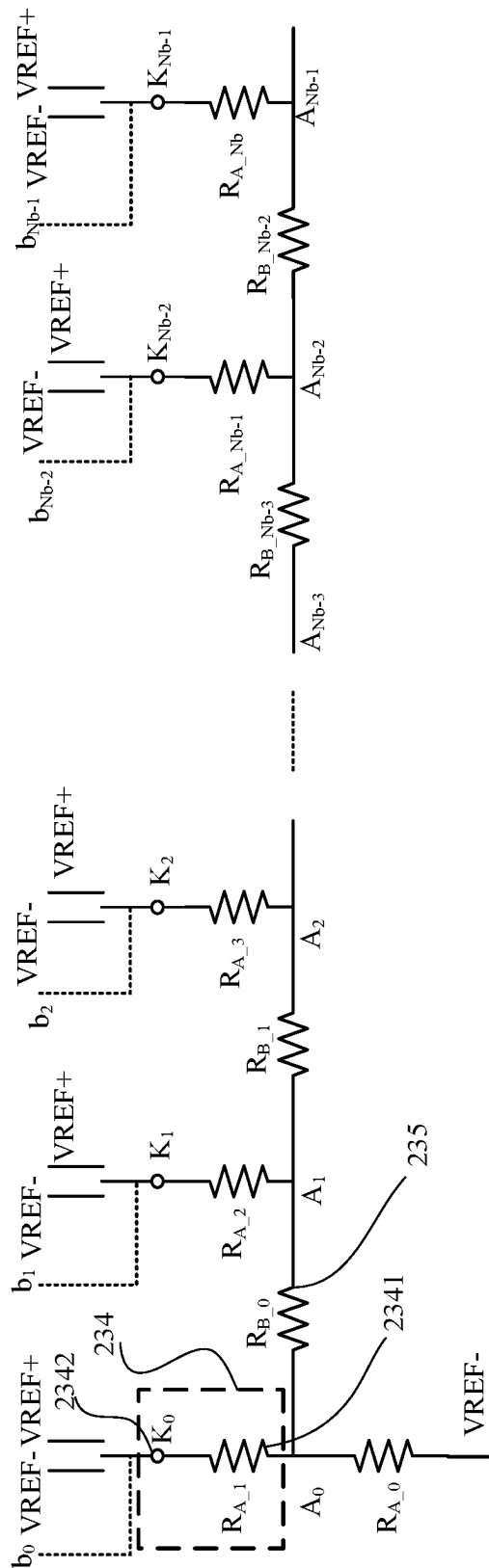


FIG. 6

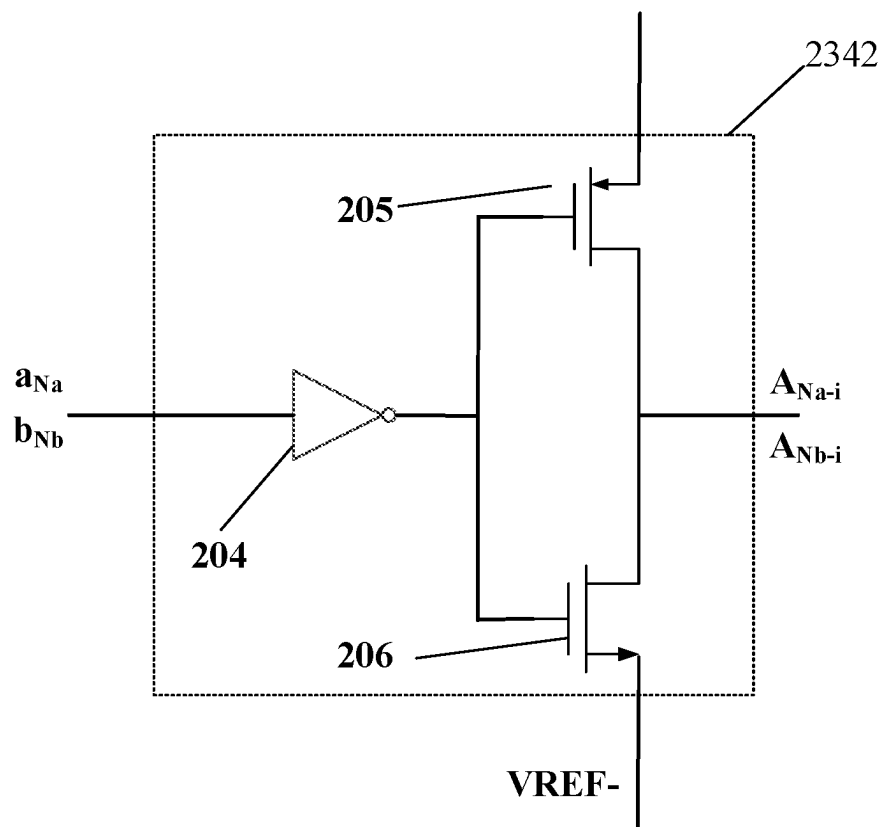


FIG. 7

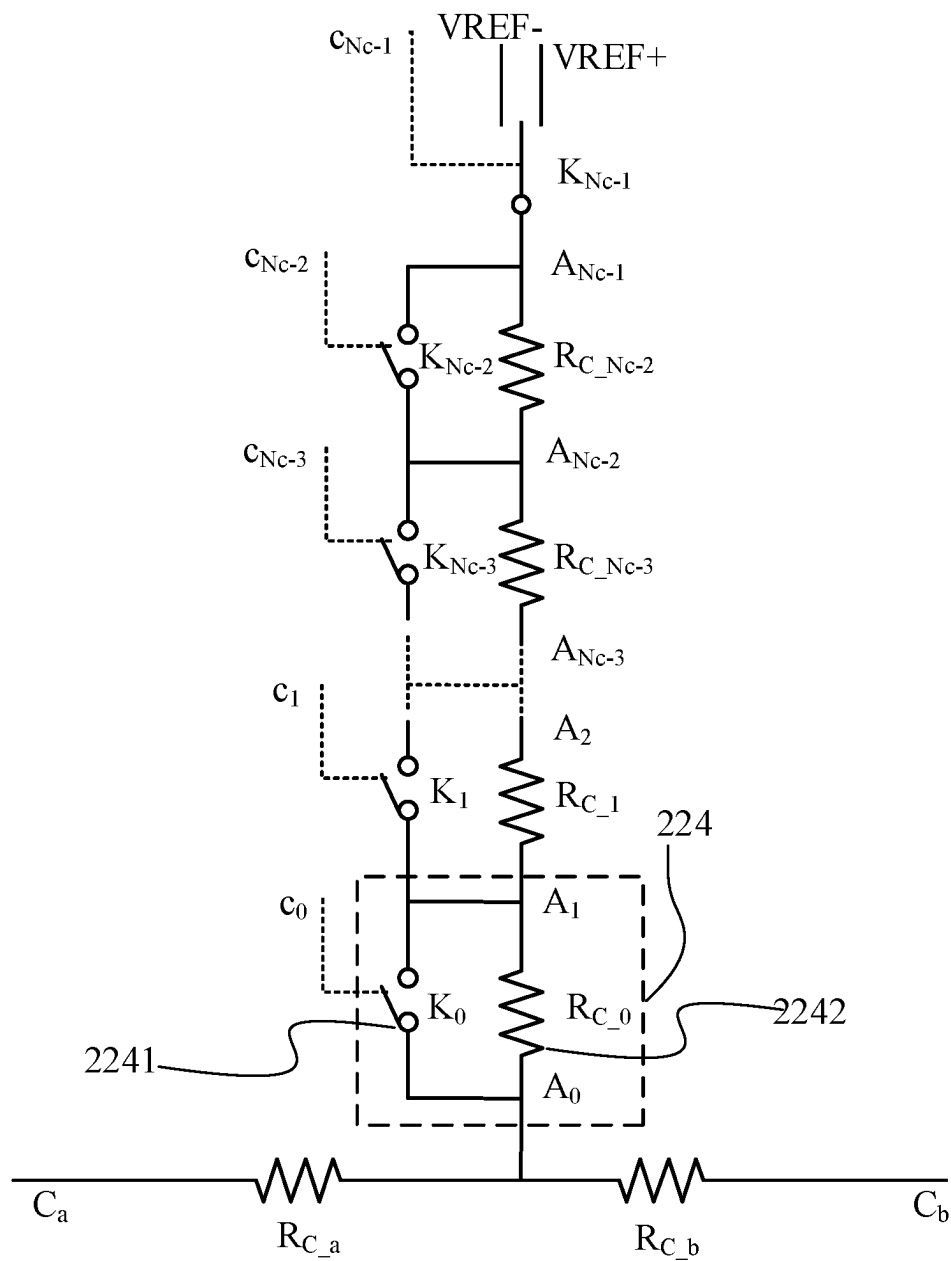


FIG. 8

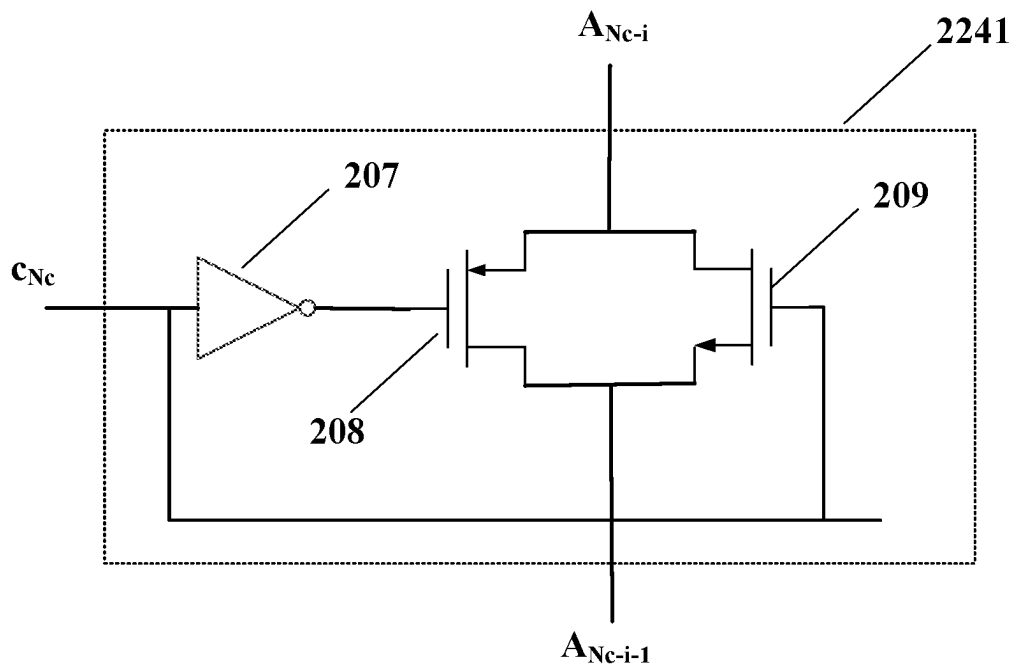


FIG. 9

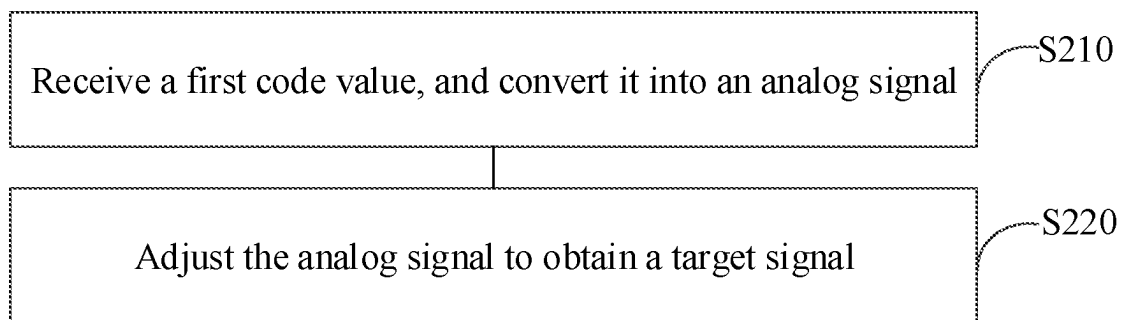


FIG. 10

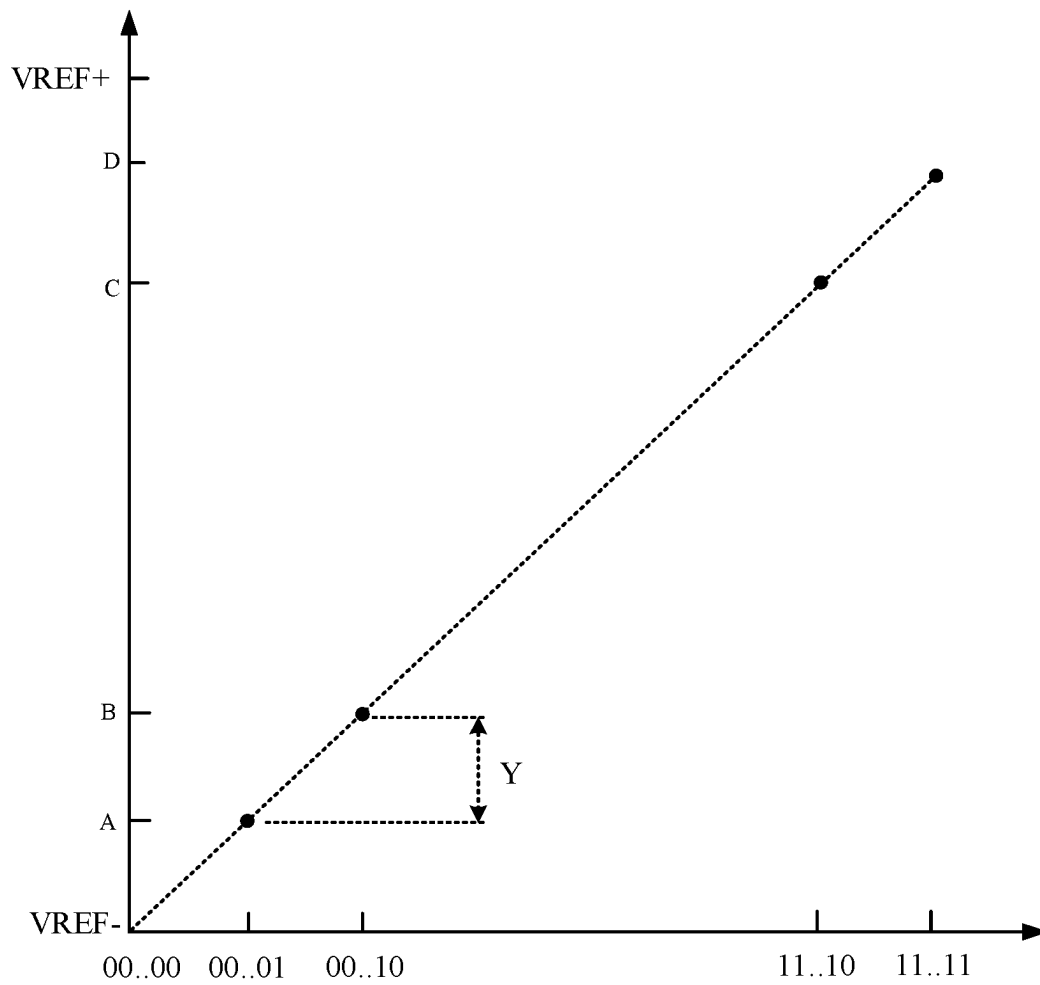


FIG. 11

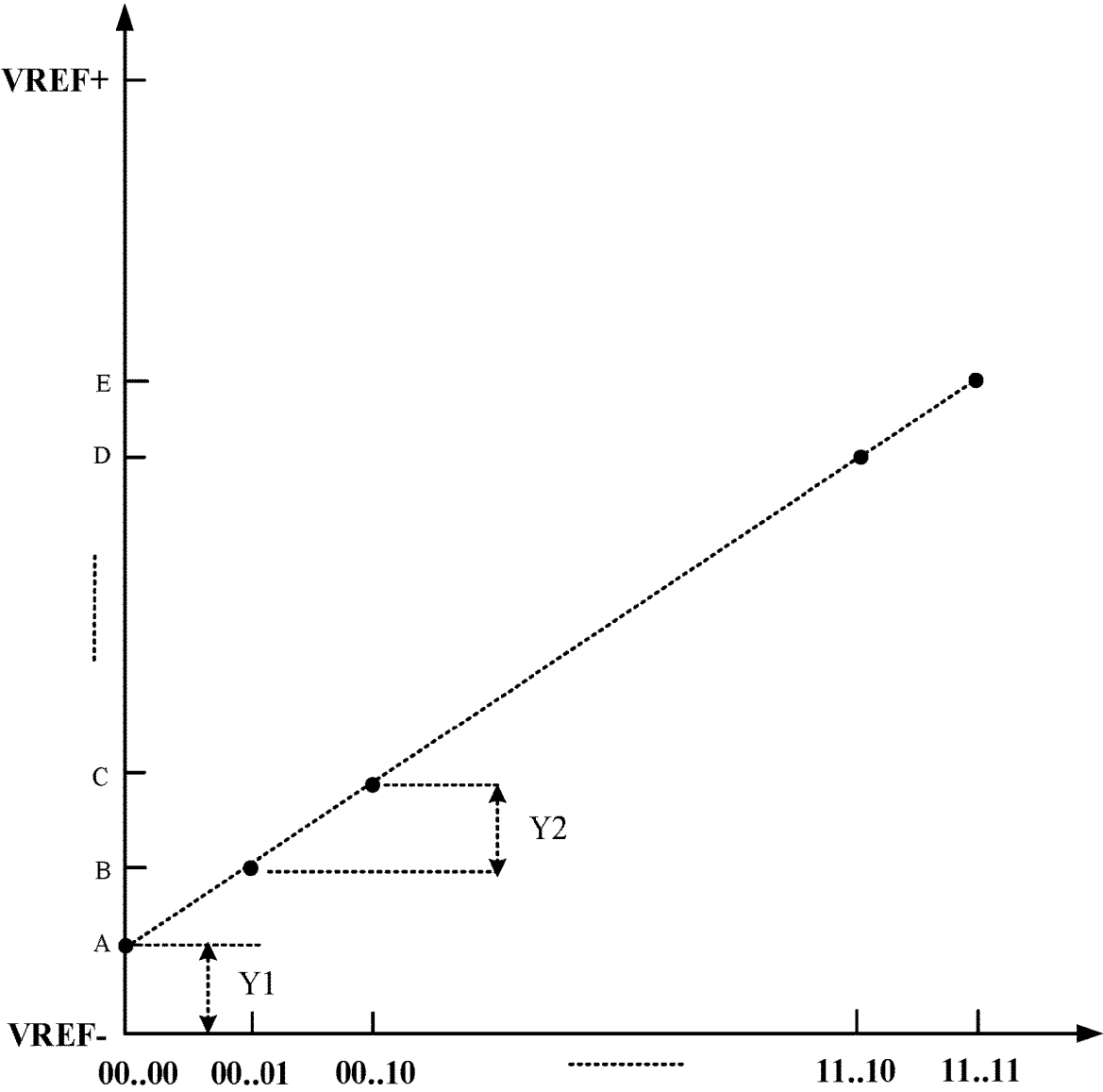


FIG. 12

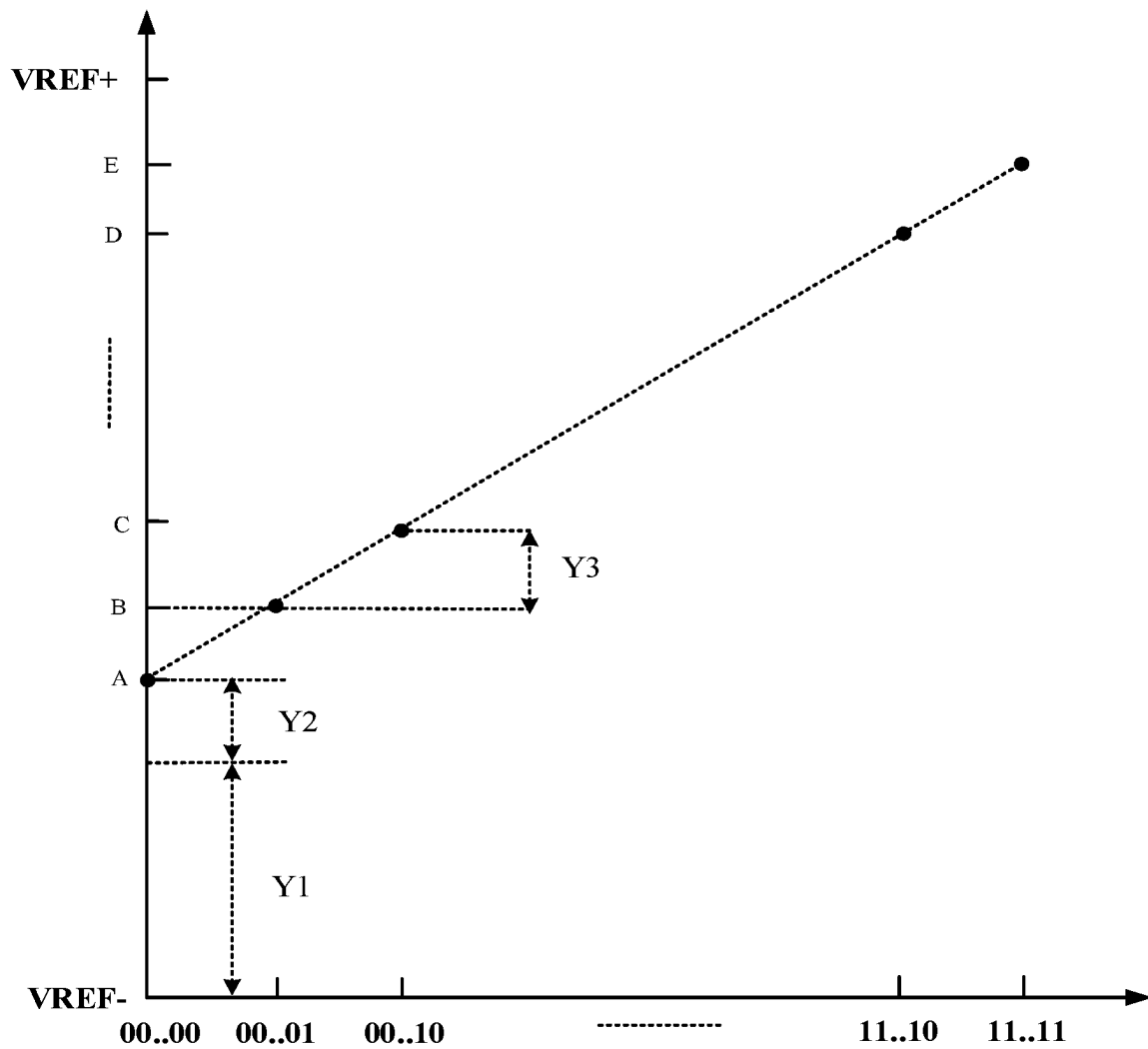


FIG. 13

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**DIGITAL-TO-ANALOG CONVERSION
CIRCUIT AND METHOD****CROSS-REFERENCING OF RELEVANT
APPLICATIONS**

The present disclosure is a continuation of International Application PCT/CN2021/079754 filed on Mar. 9, 2021, which claims priority to a Chinese Patent Application No. 202011627453.9 filed on Dec. 31, 2020. The entire contents of these applications are incorporated herein by reference.

TECHNICAL FIELD

Embodiments of the present disclosure relate to the field of integrated circuit design, and in particular to, but is not limited to, a digital-to-analog conversion circuit and method.

BACKGROUND

A Digital-to-Analog converter, also referred to as D/A or DAC, is mainly used to convert a digital signal or a discrete signal into an analog signal. The digital signal is a set of parallel encoded signals generated by a digital signal system. The digital-to-analog converter converts the encoded signal generated by the digital signal system into an equivalent analog signal, under the action of a reference voltage. And the converted analog signal enters an analog signal system after being filtered and amplified. Therefore, how to enable the digital-to-analog converter to convert a digital signal into an accurate analog signal is an urgent problem to be solved.

SUMMARY

In a first aspect, the embodiments of the present disclosure provide a digital-to-analog conversion circuit. The digital-to-analog conversion circuit includes a conversion unit and a first adjustment unit. The conversion unit includes a first code value receiving terminal used to receive a first code value, and the conversion unit is used to convert the first code value into an analog signal. The first adjustment unit includes a second code value receiving terminal used to receive a second code value. A signal input terminal of the first adjustment unit is connected to a signal output terminal of the conversion unit. The first adjustment unit is used to obtain a first analog adjustment signal according to the second code value, and adjust, with the first analog adjustment signal, the analog signal transmitted from the conversion unit, to obtain a target signal.

In a second aspect, the embodiments of the present disclosure further provide a digital-to-analog conversion method, where the method is applied to the digital-to-analog conversion circuit in the first aspect. The method includes: receiving a first code value and converting the first code value into an analog signal; and adjusting the analog signal to obtain a target signal.

Other features and corresponding beneficial effects of the present disclosure are illustrated in the following part of the specification. It is understandable that at least part of the beneficial effects become apparent from the description of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic structural diagram of a digital-to-analog conversion circuit according to an embodiment of the present disclosure;

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FIG. 2 is a schematic structural diagram of a digital-to-analog conversion circuit according to another embodiment of the present disclosure;

FIG. 3 is a schematic structural diagram of a load in the digital-to-analog conversion circuit according to another embodiment of the present disclosure;

FIG. 4 is a schematic structural diagram of a conversion unit in the digital-to-analog conversion circuit according to another embodiment of the present disclosure;

FIG. 5 is a schematic structural diagram of a second switch element in the digital-to-analog conversion circuit according to another embodiment of the present disclosure;

FIG. 6 is a schematic structural diagram of a second adjustment unit in the digital-to-analog conversion circuit according to another embodiment of the present disclosure;

FIG. 7 is a schematic structural diagram of a first switch element in the digital-to-analog conversion circuit according to another embodiment of the present disclosure;

FIG. 8 is a schematic structural diagram of a first adjustment unit in the digital-to-analog conversion circuit according to another embodiment of the present disclosure;

FIG. 9 is a schematic structural diagram of a third switch element in the digital-to-analog conversion circuit according to another embodiment of the present disclosure;

FIG. 10 is a method flowchart of a digital-to-analog conversion method according to an embodiment of the present disclosure;

FIG. 11 is a schematic diagram illustrating a relationship between digital signals and analog signals in the digital-to-analog conversion method according to an embodiment of the present disclosure;

FIG. 12 is another schematic diagram illustrating the relationship between digital signals and analog signals in the digital-to-analog conversion method according to an embodiment of the present disclosure; and

FIG. 13 is a further schematic diagram illustrating the relationship between digital signals and analog signals in the digital-to-analog conversion method according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

To illustrate the technical schemes in the embodiments of the present disclosure clearly, the drawings used in the description of the embodiments are briefly introduced below. Apparently, the drawings in the following description are merely some embodiments of the present disclosure. For those skilled in the art, other drawings may be obtained according to these drawings without paying any creative effort.

The existing digital-to-analog converters may be classified into current proportional type, voltage proportional type, and charge proportional type digital-to-analog converters. The current proportional type digital-to-analog converter sums up accurate current sources in a simple way or using binary weights. The current proportional type digital-to-analog converter has advantages of fast speed and large drive, but these advantages bring disadvantages, such as a large range of values and poor matching of elements. In the charge proportional type digital-to-analog converter, an array of parallel capacitors with binary weights are utilized, and a reference voltage is divided. The charge proportional type digital-to-analog converter has the advantages of fast speed and high accuracy, but also has the disadvantage of a large range of values of the elements. In the voltage proportional type digital-to-analog converter, voltage division through series resistors or parallel resistors are utilized, and

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the voltages are summed. The voltage proportional type digital-to-analog converter has advantages of simple structure and monotonicity, but these advantages also bring disadvantages of large area and low speed. In summary, the three types of digital-to-analog converters, namely the current proportional type, the voltage proportional type, and charge proportional type, all have a characteristic of a single and non-adjustable output voltage range, and all of them have some limitations when performing the digital-to-analog conversion.

In view of the above issues, the inventor proposes the digital-to-analog conversion circuit and method according to the embodiments of the present disclosure, which mainly aim at a technical problem of how to simply and effectively adjust the range of an analog voltage signal obtained through digital-to-analog conversion. In the embodiments of the present disclosure, by introducing a first adjustment unit, the range of a voltage obtained through digital-to-analog conversion can be simply and effectively adjusted.

In an embodiment, a digital-to-analog conversion circuit includes a conversion unit and a first adjustment unit. The conversion unit includes a first code value receiving terminal used to receive a first code value, and the conversion unit is used to convert the first code value into an analog signal. The first adjustment unit includes a second code value receiving terminal used to receive a second code value. A first signal input terminal of the first adjustment unit is connected to a signal output terminal of the conversion unit. The first adjustment unit is used to obtain a first analog adjustment signal according to the second code value, and adjust, with the first analog adjustment signal, the analog signal transmitted from the conversion unit, to obtain a target signal.

In an implementation, the digital-to-analog conversion circuit further includes a second adjustment unit. The second adjustment unit includes a third code value receiving terminal used to receive a third code value. A signal output terminal of the second adjustment unit is connected to a second signal input terminal of the first adjustment unit. The second adjustment unit is used to obtain a second analog signal according to the third code value, and adjust, with the second analog signal, the analog signal transmitted by the conversion unit.

In an implementation, the second adjustment unit includes a first resistor-selection sub-unit, a first resistor is provided between every two adjacent first resistor-selection sub-units, and the first resistor is electrically connected to its adjacent first resistor-selection sub-units.

In an implementation, each of the first resistor-selection sub-units includes a second resistor and a first switch element. A control terminal of the first switch element is used to receive the third code value, a second connection terminal of the first switch element is connected to a first connection terminal of the second resistor, and a second connection terminal of the second resistor is connected to the first resistor.

In an implementation, the first switch element includes a first inverter, a first PMOS transistor, and a first NMOS transistor. An input terminal of the first inverter is used to receive the third code value, and an output terminal of the first inverter is connected to each of a gate of the first PMOS transistor and a gate of the first NMOS transistor. A source of the first PMOS transistor is connected to a positive pole terminal of a reference voltage, a drain of the first PMOS transistor is connected to a drain of the first NMOS transistor, and a source of the first NMOS transistor is connected to a negative pole terminal of the reference voltage.

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In an implementation, the conversion unit includes a second resistor-selection sub-unit, a third resistor is provided between every two adjacent second resistor-selection sub-units, and the third resistor is electrically connected to its adjacent second resistor-selection sub-units.

In an implementation, each of the second resistor-selection sub-units includes a fourth resistor and a second switch element. A control terminal of the second switch element is used to receive the first code value, a second connection terminal of the second switch element is connected to a first connection terminal of the fourth resistor, and a second connection terminal of the fourth resistor is connected to the third resistor.

In an implementation, the second switch element includes a second inverter, a second PMOS transistor, and a second NMOS transistor. An input terminal of the second inverter is used to receive the first code value, and an output terminal of the second inverter is connected to each of a gate of the second PMOS transistor and a gate of the second NMOS transistor. A source of the second PMOS transistor is connected to the positive pole terminal of the reference voltage, a drain of the second PMOS transistor is connected to a drain of the second NMOS transistor, and a source of the second NMOS transistor is connected to the negative pole terminal of the reference voltage.

In an implementation, the first adjustment unit includes a third resistor-selection sub-unit. Each of the third resistor-selection sub-unit includes a third switch element and a fifth resistor, and the third switch element is connected to the fifth resistor.

In an implementation, the third switch element includes a third inverter, a third PMOS transistor, and a third NMOS transistor. An input terminal of the third inverter is used to receive the second code value, an output terminal of the third inverter is connected to a gate of the third PMOS transistor, a source of the third PMOS transistor is connected to a drain of the third NMOS transistor, and a drain of the third PMOS transistor is connected to a source of the third NMOS transistor.

In an embodiment, a digital-to-analog conversion method is provided, where the method is applied to the digital-to-analog conversion circuit mentioned above. The method includes: receiving a first code value and converting the first code value into an analog signal; and adjusting the analog signal to obtain a target signal.

In an implementation, the adjusting the analog signal to obtain a target signal includes: obtaining a digital-to-analog adjustment list, and obtaining, based on the digital-to-analog adjustment list, the target signal corresponding to the analog signal. The digital-to-analog adjustment list is obtained by obtaining multiple full-code-value digital signals and adjusting analog signals corresponding to the multiple full-code-value digital signals.

In an implementation, the adjusting, by the first adjustment unit, the analog signal to obtain a target signal, includes: receiving, by the first adjustment unit, a second code value; obtaining a first analog adjustment signal according to the second code value; and adjusting, with the first analog adjustment signal, the analog signal obtained from the conversion unit, to obtain a target signal.

In an implementation, the digital-to-analog conversion circuit further includes a second adjustment unit, and the method further includes: receiving, by the second adjustment unit, a third code value; obtaining a second analog signal according to the third code value; and adjusting, with the second analog signal, the analog signal obtained by the conversion unit.

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In an implementation, the second adjustment unit includes n first resistor-selection sub-units, and a first resistor is electrically connected between every two adjacent first resistor-selection sub-units. Each of the first resistor-selection sub-units includes a second resistor and a first switch element, a control terminal of the first switch element is configured to receive a bit of the third code value, a second connection terminal of the first switch element is connected to a first connection terminal of the second resistor, and a second connection terminal of the second resistor is connected to the first resistor. The method further includes: in response to detecting that the received bit is 0, controlling the first switch element to be turned off and connected to a negative pole terminal of a reference voltage; and in response to detecting that the received bit is 1, controlling the first switch element to be turned on and connected to a positive pole terminal of the reference voltage.

In an implementation, the conversion unit includes m second resistor-selection sub-units, and a third resistor is electrically connected between every two adjacent second resistor-selection sub-units. Each of the second resistor-selection sub-units includes a fourth resistor and a second switch element, a control terminal of the second switch element is configured to receive a bit of the first code value, a second connection terminal of the second switch element is connected to a first connection terminal of the fourth resistor, and a second connection terminal of the fourth resistor is connected to the third resistor. The method further includes: in response to detecting that the received bit is 0, controlling the second switch element to be turned off and connected to a negative pole terminal of a reference voltage, and in response to detecting that the received bit is 1, controlling the second switch element to be turned on and connected to a positive pole terminal of a reference voltage.

In an implementation, the first adjustment unit includes k third resistor-selection sub-units, each of the third resistor-selection sub-units includes a third switch element and a fifth resistor connected in parallel to each other, and a control terminal of each third switch element is configured to receive a bit of the second code value. The method further includes: in response to detecting that the received bit is 0, controlling the third switch element to be turned off; and in response to detecting that the received bit is 1, controlling the third switch element to be turned on, to make the fifth resistor connected in parallel to the third switch element be short circuited.

In the digital-to-analog conversion circuit provided by the embodiments of the present disclosure, a conversion unit and a first adjustment unit are incorporated to simply and effectively adjust a voltage range. Specifically, the digital-to-analog conversion circuit may include the conversion unit and the first adjustment unit. Among them, the conversion unit may include a first code value receiving terminal used to receive a first code value, and the conversion unit is used to convert the first code value into an analog signal. The first adjustment unit may include a second code value receiving terminal used to receive a second code value, a signal input terminal of the first adjustment unit is connected to a signal output terminal of the conversion unit, and the first adjustment unit is used to obtain a first analog adjustment signal based on the second code value, and adjust, with first analog adjustment signal, the analog signal transmitted from the conversion unit, to obtain a target signal. In the present disclosure, after the conversion unit converts the received first code value into the analog signal, the analog signal may be adjusted by the first adjustment unit, which can make the finally obtained target signal accurate and effective.

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Referring to FIG. 1, a schematic structural diagram of a digital-to-analog conversion circuit according to an embodiment of the present disclosure is illustrated. As illustrated in FIG. 1, the digital-to-analog conversion circuit 100 may include a conversion unit 110 and a first adjustment unit 120.

In an implementation, the conversion unit 110 may include a first code value receiving terminal 111. The first code value receiving terminal 111 is used to receive a first code value. The conversion unit 110 is used to convert the first code value into an analog signal. The N_a bits illustrated in FIG. 1 may be referred to as the first code value, and V_{dac_a} is the converted analog signal which may also be referred to as an output voltage of the conversion unit. The first code value may be a set of parallel coded signals generated by a digital signal system, and it may be a binary code value. For example, the first code value may be a binary code value of two bits, a binary code value of four bits, a binary code value of eight bits, or a binary code value of sixteen bits. The specific number of the bits is not explicitly limited here, and it may be chosen based on actual situations. In addition, the first code value may also be a binary code value input by the user according to his/her needs. For example, when the user wants to convert a digital signal 011 into an analog signal, he/she may input 011 into the conversion unit 110, and obtain the converted analog signal through the conversion unit 110.

In the embodiment of the present disclosure, the conversion unit 110 may be constituted by multiple switches and unit resistors, and the number of the switches and the number of the resistors are determined by the first code value. The first code value ranges from 0 to $2^{N_a}-1$, and 2^{N_a} analog output voltages may be obtained, where N_a may be the number of the binary bits. For example, when the first code value is 1011, the corresponding N_a is 4.

In addition, as illustrated in FIG. 1, besides the first code value receiving terminal 111, the conversion unit 110 may further include a signal output terminal 112, as well as a positive pole terminal and a negative pole terminal of a reference voltage. The positive pole terminal of the reference voltage is V_{REF+} illustrated in FIG. 1, and the negative pole terminal of the reference voltage is V_{REF-} illustrated in FIG. 1. In the embodiments of the present disclosure, the positive pole terminal of the reference voltage and the negative pole terminal of the reference voltage have a certain ability of driving, that is, the positive pole terminal of the reference voltage and the negative pole terminal of the reference voltage may provide an output current. The range of the voltage of analog signal output by the conversion unit 110 is determined by the range between V_{REF+} and V_{REF-} of the reference voltage, that is $(V_{REF+})-(V_{REF-})$.

In some embodiments, the first code value generated by the digital signal system may be constituted by m_a "0" and n_a "1", where $m_a+n_a=N_a$. When there is a "0" in the first code value, a corresponding internal switch of the conversion unit 110 is connected to the negative pole terminal of the reference voltage V_{REF-} ; and when there is a "1" in the first code value, a corresponding internal switch of the conversion unit 110 is connected to the positive pole terminal of the reference voltage V_{REF+} . Through the switches, a range of series or parallel resistors exist between V_{REF+} and V_{REF-} , and an analog voltage signal proportional to $(V_{REF+})-(V_{REF-})$ is obtained at the signal output terminal 112.

In an implementation, when $m_a=N_a$ and $n_a=0$, the output of the conversion unit 110 is minimum, and all the bits of the first code value in this case are 0. For example, when $m_a=4$ and $n_a=0$, the corresponding first code value is 0000. In

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addition, when $m_a=0$ and $n_a=N_a$, the output of the conversion unit **110** is maximum, and all the bits of the first code value in this case are 1. For example, when $m_a=0$ and $n_a=4$, the corresponding first code value is 1111. The lowest significant bit (LSB) of the conversion unit **110** is the lowest bit of the N_a bits, and the minimum variation in output corresponding to the lowest bit is

$$1LSB = \frac{V_{REF+} + V_{REF-}}{2^{N_a}}.$$

The maximum analog output voltage is a full-scale voltage

$$V_{FS_a} = \frac{2^{N_a} - 1}{2^{N_a}} (V_{REF+} + V_{REF-}).$$

In the embodiment of the present disclosure, the first adjustment unit **120** may include a second code value receiving terminal **121**. The second code value receiving terminal **121** is used to receive a second code value. A signal input terminal **122** of the first adjustment unit **120** is connected to the signal output terminal **112** of the conversion unit **110**. The first adjustment unit **120** is used to obtain a first analog adjustment signal based on the second code value, and adjust, with the first analog adjustment signal, the analog signal transmitted from the conversion unit **110**, to obtain a target signal.

As illustrated in FIG. 1, the N_c bits may be referred to as the second code value. The second code value may also be a set of parallel coded signals generated by the digital signal system, and it may be a binary code value. For example, the second code value may be a binary code value of two bits, a binary code value of eight bits, or a sixteen-bit binary code value of sixteen bits, where the specific number of the bits is not specifically limited here, and it may be chosen based on the actual situations. In addition, the second code value may also be a binary code value input by the user according to his/her needs.

In the embodiment of the present disclosure, the first adjustment unit **120** may be constituted by multiple switches and unit resistors. As can be seen from FIG. 1, besides the second code value receiving terminal **121**, the first adjustment unit **120** may further include a signal input terminal **122**, as well as a positive pole terminal and a negative pole terminal of a reference voltage. As illustrated in FIG. 1, the positive pole terminal of the reference voltage is V_{REF+} , and the negative pole terminal of the reference voltage is V_{REF-} .

In an implementation, a first input of the first adjustment unit **120** is the binary code value of the N_c bits, and the binary code value of the N_c bits is the second code value. The highest bit of the second code value (the code value of N_c bits) is used to select the output range of the conversion unit **110**, and the remaining (N_c-1) bits are used to adjust the value of the LSB voltage of the conversion unit **110**. By adjusting the (N_c-1) bits, 2^{N_c-1} values of the lowest significant bit (LSB) voltage may be obtained.

In another implementation, other two inputs of the first adjustment unit **120** are from the positive pole terminal V_{REF+} and the negative pole terminal V_{REF-} of the reference voltage. In addition, the signal input terminal **122** of the first adjustment unit **120** may be used to input the output voltage of the conversion unit **110**. The analog signal output by the conversion unit **110** may be processed by the first adjustment unit **120** with the N_c bits, to output an analog voltage capable

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of being flexibly adjusted. The output voltage of the first adjustment unit **120** is used to drive a capacitive load, or drive, through a buffer circuit, a resistive load and a capacitive load which has a specific speed requirement.

In the digital-to-analog conversion circuit provided by the embodiment of the present disclosure, a conversion unit and a first adjustment unit are incorporated to simply and effectively adjust a voltage range. Specifically, the digital-to-analog conversion circuit may include the conversion unit and the first adjustment unit. Among them, the conversion unit may include a first code value receiving terminal used to receive a first code value, and the conversion unit is used to convert the first code value into an analog signal. The first adjustment unit may include a second code value receiving terminal used to receive a second code value, a signal input terminal of the first adjustment unit is connected to a signal output terminal of the conversion unit, and the first adjustment unit is used to obtain a first analog adjustment signal based on the second code value, and adjust, with first analog adjustment signal, the analog signal transmitted from the conversion unit, to obtain a target signal. In the present disclosure, after the conversion unit converts the received first code value into the analog signal, the analog signal may be adjusted by the first adjustment unit, which can make the finally obtained target signal accurate and effective.

Referring to FIG. 2, which is a schematic structural diagram of a digital-to-analog conversion circuit according to another embodiment of the present disclosure. Besides the conversion unit **210** and the first adjustment unit **220**, the digital-to-analog conversion circuit **200** may further include a second adjustment unit **230**.

In some embodiments, the second adjustment unit **220** also includes a third code value receiving terminal **231**, and the third code value receiving terminal **231** is used to receive a third code value. A signal output terminal **232** of the second adjustment unit **230** may be connected with the signal input terminal **223** of the first adjustment unit **220**. The second adjustment unit **230** is used to obtain a second analog signal according to the third code value, and adjust, with the second analog signal, the analog signal transmitted by the conversion unit **210**.

In an implementation, the N_b bits illustrated in FIG. 2 may be referred to as the third code value, and V_{dac_b} is the converted second analog signal which may also be referred to as the output voltage signal of the second adjustment unit. The third code value may be a set of parallel coded signals generated by the digital signal system, and it may be a binary code value. For example, the first code value may be a binary code value of two bits, a binary code value of four bits, a binary code value of eight bits, or a binary code value of sixteen bits. The specific number of the bits is not explicitly limited here, and it may be chosen based on the actual situations.

In an implementation, the second adjustment unit **230** may be constituted by multiple switches and unit resistors, and the number of the switches and the number of the resistors are determined by the third code value. The third code value ranges from 0 to $2^{N_b}-1$ and 2^{N_b} analog output voltages may be obtained, where N_b may be the number of the binary bits. In addition, as can be seen from FIG. 2, besides the third code value receiving terminal **231**, the second adjustment unit may further include a signal output terminal **232**, as well as a positive pole terminal and a negative pole terminal of a reference voltage. The positive pole terminal of the reference voltage is V_{REF+} illustrated in FIG. 2, and the negative pole terminal of the reference voltage is V_{REF-} illustrated in FIG. 2.

In some embodiments, the third code value generated by the digital signal system may be constituted by m_b "0" and n_b "1", where $m_b + n_b = N_b$. When there is a "0" in the third code value, a corresponding internal switch of the second adjustment unit **230** is connected to the negative pole terminal of the reference voltage V_{REF-} ; and when there is a "1" in the third code value, a corresponding internal switch of the second adjustment unit **230** is connected to the positive pole terminal of the reference voltage V_{REF+} . In the embodiment of the present disclosure, through the series and parallel resistors, a second analog voltage signal, which is in a certain proportion to $(V_{REF+}) - (V_{REF-})$, may be obtained through the second adjustment unit **230**.

In an implementation, when $m_b = N_b$ and $n_b = 0$, the output of the second adjustment unit **230** is minimum, and all the bits of the third code value in this case are 0. For example, when $m_b = 3$ and $n_b = 0$, the corresponding third code value is 000. In addition, when $m_b = 0$ and $n_b = N_b$, the output of the second adjustment unit **230** is maximum, and all the bits of the third code value in this case are 1. For example, when $m_b = 0$ and $n_b = 3$, the corresponding third code value is 111. The lowest significant bit (LSB) of the second adjustment unit **230** is the lowest bit of the N_b bits, and the minimum variation in output corresponding to the lowest bit is

$$1LSB = \frac{V_{REF+} - V_{REF-}}{2^{N_b}}.$$

The maximum analog output voltage is a full-scale voltage

$$V_{FS,b} = \frac{2^{N_b} - 1}{2^{N_b}} (V_{REF+} + V_{REF-}).$$

In the embodiment of the present disclosure, the first adjustment unit **220** may be constituted by multiple switches and unit resistors. The first adjustment unit **220** may be referred to as a summation circuit. A first input of the first adjustment unit **220** is the binary code value of the N_c bits, and the binary code value of the N_c bits is the second code value. The highest bit of the second code value (the code value of N_c bits) is used to select the output range of the conversion unit **110**, and the remaining $(N_c - 1)$ bits are used to adjust the value of the LSB voltage of the conversion unit **210**. $2^{N_c - 1}$ values of the lowest significant bit (LSB) voltage may be obtained, by adjusting the $(N_c - 1)$ bits.

In another implementation, a second input and a third input of the first adjustment unit **220** are from the positive pole terminal V_{REF+} and the negative pole terminal V_{REF-} of the reference voltage, a fourth input of the first adjustment unit **220** is the output voltage of the conversion unit **210**, and a fifth input of the first adjustment unit **220** is the output voltage of the second adjustment unit **230**. After the analog signal output by the conversion unit **210** and the analog signal output by the second adjustment unit **230** are processed by the first adjustment unit **220**, an analog voltage capable of being flexibly adjusted is output. The output voltage of the first adjustment unit **220** is used to drive a capacitive load, or drive, through a buffer circuit, a resistive load and a capacitive load which has a specific speed requirement.

In some other embodiments, the digital-to-analog conversion circuit **200** may include a load **240**. The load **240** is connected to the first adjustment unit **220**, and receives the target signal (V_{dac}) transmitted from the first adjustment unit

220. In other words, the target signal obtained from the first adjustment unit **220** is mainly used to power the load **240**. Referring to FIG. 3, the load **240** may include a resistor R_L and a capacitor C_L . A first connection terminal of the resistor R_L is connected to the first adjustment unit **220** through a node D_0 . A second connection terminal of the resistor R_L is connected to a node D_1 , and the node D_1 is connected to a first connection terminal of the capacitor C_L . A second connection terminal of the capacitor C_L is grounded (GND). The load **240** may be determined by a subordinate circuit. When a large capacitive or resistive load needs to be driven, in the embodiment of the present disclosure, a buffer circuit may be inserted in the load **240**.

It is notable that, the conversion unit **210** in the digital-to-analog conversion circuit **200** may serve as a main circuit for output, the second adjustment unit **230** may serve as a slave circuit for output, and the first adjustment unit **220** serves as a control and adjustment circuit. When the conversion unit **210** outputs the analog voltage with a specific range, in the embodiment of the present disclosure, the second adjustment unit **230** may be controlled to adjust a common mode voltage output by the conversion unit **210**, where the common mode voltage may be applied with $(2^{N_b} - 1)$ adjustments. The first adjustment unit **210** may simultaneously adjust the minimum variation in the LSB of the conversion unit **210** and the second adjustment unit **230**. In another implementation, in the embodiments of the present disclosure, the conversion unit **210** may be adjusted to serve as the slave circuit for output, the second adjustment unit **230** serves as the main circuit for output, and the control role of the first adjustment unit **220** is unchanged.

To sum up, the conversion unit **210** is used to convert the first code value (the binary code value of N_a bits) into an analog voltage in a specific proportion to a voltage defined by the positive and negative reference voltages. The second adjustment unit **230** is used to convert the third code value (the binary code value of N_b bits) into a second analog voltage in a specific proportion to a voltage defined by the positive and negative reference voltages. The first adjustment unit **220** sums up the analog output of the conversion unit **210** and the analog output of the second adjustment unit **230**, to obtain the final analog output voltage, i.e., the target signal. The load **240** is the output driving stage of the overall analog voltage, and the load **240** usually is a capacitive load.

As can be seen from the above introduction, the conversion unit **210** may include the switches and the unit resistors, and the input of the conversion unit **210** includes the first code value (a code value signal of N_a bits) generated by the digital signal system, and those from the positive pole terminal V_{REF+} and the negative pole terminal V_{REF-} of the reference voltage. The "1" or "0" in the first code value (the code value signal of N_a bits) enables the corresponding switch to be controlled to be turned on or off, and makes it connected to V_{REF+} or the V_{REF-} . After the voltage division by the internal resistors, an analog signal $V_{dac,a}$ in one-to-one correspondence to the code value is finally obtained.

Referring to FIG. 4, the conversion unit may include m second resistor-selection sub-units **213**, and a third resistor **214** is provided between every two adjacent second resistor-selection sub-units **213**. The third resistor **214** is connected to its adjacent second resistor-selection sub-units **213**. In addition, each second resistor-selection sub-unit includes a fourth resistor **2131**, and a second switch element **2132**. A control terminal of the second switch element **2132** is used to receive the first code value, a second connection terminal of the second switch element **2132** is connected to a first

connection terminal of the fourth resistor **2131**, and a second connection terminal of the fourth resistor **2131** is connected to the third resistor **214**.

In addition, as can be seen from FIG. 4, the conversion unit **210** may include single-pole double-throw switches $K_0, K_1, \dots, K_{Na-2}$, and K_{Na-1} , resistors $R_{A_0}, R_{A_1}, \dots, R_{A_{Na-1}}$ and $R_{A_{Na}}$, and resistors $R_{B_0}, R_{B_1}, \dots, R_{B_{Na-3}}$ and $R_{B_{Na-2}}$. Among others, $R_{A_i} = 2R$ ($i=0, 1, \dots, N_{a-1}, Na$), $R_{B_i} = R$ ($i=0, 1, \dots, N_{a-3}, N_{a-2}$), and the R is unit resistance. The N_a bits of the binary code $a_{Na-1}a_{Na-2} \dots a_2a_1a_0$ control selection of the corresponding switches, respectively. When one bit of $a_{Na-1}a_{Na-2} \dots a_2a_1a_0$ is "0", its corresponding switch of $K_0, K_1, \dots, K_{Na-2}$ and K_{Na-1} is connected to V_{REF-} . When one bit of $a_{Na-1}a_{Na-2} \dots a_2a_1a_0$ is "1", its corresponding switch of $K_0, K_1, \dots, K_{Na-2}$ and K_{Na-1} is connected to V_{REF+} . Switch k_0 is a three-terminal device controlled by a_0 in the N_a code value, with the first terminal of switch k_0 connected to V_{REF-} , the second terminal connected to V_{REF+} , and the third terminal connected to the first terminal of resistor R_{A_1} . The second terminal of resistor R_{A_1} is connected to node A_0 , and node A_0 is in turn connected to each of the first terminal of resistor R_{A_0} and the first terminal of resistor R_{B_0} . The second terminal of resistor R_{A_0} is connected to V_{REF-} . The second terminal of resistor R_{B_0} is connected to node A_1 . Further, node A_1 is connected to the first terminal of resistor R_{A_2} and the first terminal of resistor R_{B_1} . The second terminal of resistor R_{A_2} is connected to the third terminal of switch K_1 . Switch K_1 is controlled by bit a_1 of the code value of N_a bits. The first terminal of switch K_1 is connected to V_{REF-} , and the second terminal of switch K_1 is connected to V_{REF+} . The second terminal of resistor R_{B_1} is connected to node A_2 . Further, node A_2 is connected to each of the first terminal of resistor R_{A_3} and the first terminal of resistor R_{B_2} . The second terminal of resistor R_{A_3} is connected to the third terminal of switch K_2 . Switch K_2 is controlled by the bit a_2 of the code value of N_a bits. The first terminal of switch K_2 is connected to V_{REF-} , and the second terminal of switch K_2 is connected to V_{REF+} . The second terminal of resistor R_{B_2} is connected to node A_3 . The subsequent connections follow the previous pattern of connection. The second terminal of resistor $R_{B_{Na-3}}$ is connected to node A_{Na-2} . Further, node A_{Na-2} is connected to each of the first terminal of resistor $R_{A_{Na-1}}$ and the first terminal of resistor $R_{B_{Na-2}}$. The second terminal of resistor $R_{A_{Na-1}}$ is connected to the third terminal of switch K_{Na-2} . Switch K_{Na-2} is controlled by the bit a_{Na-2} of the code value of N_a bits. The first terminal of switch K_{Na-2} is connected to V_{REF-} , and the second terminal of switch K_{Na-2} is connected to V_{REF+} . The second terminal of resistor $R_{B_{Na-2}}$ is connected to node A_{Na-1} . Further, node A_{Na-1} is connected to each of resistor $R_{A_{Na}}$ and the first terminal of the summation circuit of next stage. Among them, the binary code of N_a bits is the first code value, and the N_{a-1} is m .

Referring to FIG. 5, the second switch element **2132** may include a second inverter **201**, a second PMOS transistor **202**, and a second NMOS transistor **203**. The input terminal of the second inverter **201** is used to receive the first code value, and the output terminal of the second inverter **201** is connected to each of the gate of the second PMOS transistor **202** and the gate of the second NMOS transistor **203**. The source of the second PMOS transistor **202** is connected to the positive pole terminal of the reference voltage, the drain of the second PMOS transistor **202** is connected to the drain of the second NMOS transistor **203**, and the source of the second NMOS transistor **203** may be connected to the negative pole terminal of the reference voltage.

In a specific implementation, the second switch element **2132** is a switch with four ports, where the drain of the second PMOS transistor **202** and the drain of the second NMOS transistor **203** are connected and they are connected to node A_i ($i=0, 1, \dots, N_{a-1}$) in FIG. 4. The source of the second NMOS transistor **203** is connected to V_{REF-} . The input of the inverter **201** is connected to a_{Na} in the code value of N_a bits.

In some other implementations, the second adjustment unit **230** may also include switches and unit resistors, and the input of the second adjustment unit **230** includes the third code value (a code value control signal of N_b bits) from the digital signal system, the positive reference voltage V_{REF+} and the negative reference voltage V_{REF-} . The "1" or "0" in the code value of N_b bits enables the corresponding switch to be turned on or off, and makes it connected to V_{REF+} or V_{REF-} . After the voltage division by the internal resistors, an analog signal $V_{dac,b}$ in one-to-one correspondence to the code value is finally obtained.

Referring to FIG. 6, the second adjustment unit **230** may include n first resistor-selection sub-units **234**, and a first resistor **235** is provided between every two adjacent first resistor-selection sub-units **234**. The first resistor **235** is electrically connected to its adjacent first resistor-selection sub-units **234**. In addition, each first resistor-selection sub-unit **234** includes a second resistor **2341**, and a first switch element **2342**. A control terminal of the first switch element **2342** is used to receive the third code value, a second connection terminal of the first switch element **2342** is connected to a first connection terminal of the second resistor **2341**, and a second connection terminal of the second resistor **2341** is connected to the first resistor **235**.

In addition, as can be seen from FIG. 6, the second adjustment unit **230** may include single-pole double-throw switches of $K_0, K_1, \dots, K_{Nb-2}$ and K_{Nb-1} , resistors $R_{A_0}, R_{A_1}, \dots, R_{A_{Nb-1}}$ and $R_{A_{Nb}}$, and resistors $R_{B_0}, R_{B_1}, \dots, R_{B_{Nb-3}}$ and $R_{B_{Nb-2}}$. Among them, $R_{A_i} = 2R$ ($i=0, 1, \dots, N_{b-1}, Nb$), $R_{B_i} = R$ ($i=0, 1, \dots, N_{b-3}, N_{b-2}$), and R is unit resistance. The N_b bits of the binary code $b_{Nb-1}b_{Nb-2} \dots b_2b_1b_0$ control selection of the corresponding switches, respectively. When one bit of $b_{Nb-1}b_{Nb-2} \dots b_2b_1b_0$ is "0", its corresponding switch of $K_0, K_1, \dots, K_{Nb-2}$ and K_{Nb-1} is connected to V_{REF-} . When one bit of $b_{Nb-1}b_{Nb-2} \dots b_2b_1b_0$ is "1", its corresponding switch of $K_0, K_1, \dots, K_{Nb-2}$ and K_{Nb-1} is connected to V_{REF+} . Switch K_0 is a three-terminal device controlled by b_0 in the N_b code value, with the first terminal of switch K_0 connected to V_{REF-} , the second terminal connected to V_{REF+} , and the third terminal connected to the first terminal of resistor R_{A_1} . The second terminal of resistor R_{A_1} is connected to node A_0 , and node A_0 is in turn connected to each of the first terminal of resistor R_{A_0} and the first terminal of resistor R_{B_0} . The second terminal of resistor R_{A_0} is connected to V_{REF-} . The second terminal of resistor R_{B_0} is connected to node A_1 . Further, node A_1 is connected to each of the first terminal of resistor R_{A_2} and the first terminal of resistor R_{B_1} . The second terminal of resistor R_{A_2} is connected to the third terminal of switch K_1 . Switch K_1 is controlled by the bit b_1 of the code value of N_b bits. The first terminal of switch K_1 is connected to V_{REF+} , and the second terminal of switch K_1 is connected to V_{REF-} . The second terminal of resistor R_{B_1} is connected to node A_2 . Further, node A_2 is connected to each of the first terminal of resistor R_{A_3} and the first terminal of resistor R_{B_2} . The second terminal of resistor R_{A_3} is connected to the third terminal of switch K_2 . Switch K_2 is controlled by the bit b_2 of the code value of N_b bits. The first terminal of switch K_2 is connected to V_{REF-} ,

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and the second terminal of switch K_2 is connected to V_{REF+} . The second terminal of resistor R_{B-2} is connected to node A_3 . The subsequent connections follow the previous pattern of connection. The second terminal of resistor R_{B-Nb-3} is connected to node A_{Nb-2} . Further, node A_{Nb-2} is connected to each of the first terminal of resistor R_{A-Nb-1} and the first terminal of resistor R_{B-Nb-2} . The second terminal of resistor R_{A-Nb-1} is connected to the third terminal of switch K_{Nb-2} . Switch K_{Nb-2} is controlled by the bit b_{Nb-2} of the code value of N_b bits. The first terminal of switch K_{Nb-2} is connected to V_{REF-} , and the second terminal thereof is connected to V_{REF+} . The second terminal of resistor R_{B-Nb-2} is connected to node A_{Nb-1} . Further, node A_{Nb-1} is connected to each of resistor R_{A-Nb} and the second terminal of the summation circuit of next stage. Among them, the binary code of N_b bits is the third code value, and the N_{b-1} is n .

Referring to FIG. 7, the first switch element **2342** may include a first inverter **204**, a first PMOS transistor **205**, and a first NMOS transistor **206**. The input terminal of the first inverter **204** is used to receive the third code value, and the output terminal of the first inverter **204** is connected to each of the gate of the first PMOS transistor **205** and the gate of the first NMOS transistor **206**. The source of the first PMOS transistor **205** is connected to the positive pole terminal of the reference voltage, the drain of the first PMOS transistor **205** is connected to the drain of the first NMOS transistor **206**, and the source of the first NMOS transistor **206** may be connected to the negative pole terminal of the reference voltage.

In a specific implementation, the first switch element **2342** may be a switch with four ports, where the drain of the first NMOS transistor **206** and the drain of the first PMOS transistor **205** are connected and they are connected to node A_i ($i=0, \dots, N_{b-1}$) in FIG. 6. The source of the first PMOS transistor **205** is connected to V_{REF-} . The input of the inverter **204** is connected to b_{Nb} in the code value of N_b bits.

In some other implementations, the first adjustment unit **220** may be constituted by switches and resistors, and the input of the first adjustment unit **220** includes the second code value (a code value control signal of N_c bits) of the digital signal system, the positive reference voltage V_{REF+} , the negative reference voltage V_{REF-} , as well as an analog output voltage of N_a bits and an analog output voltage of N_b bits. The "1" or "0" in the code value of N_c bits enables its corresponding switch to be turned on or off, and makes it connected to the respective V_{REF+} or V_{REF-} . After the first adjustment unit **220** performs a summarizing operation on the two analog voltages, a final target signal V_{dac} is output, and the target signal may be an analog voltage.

Referring to FIG. 8, the first adjustment unit **220** may include k third resistor-selection sub-units **224**. Each third resistor-selection sub-unit may include a third switch element **2241** and a fifth resistor **2242**, and the third switch element **2241** is connected parallel to the fifth resistor **2242**. As can be seen from FIG. 8, the first adjustment unit **220** may include single-pole single-throw switches $K_0, K_1, \dots, K_{Nc-2}$, a single-pole single-throw switch K_{Nc-1} , resistors $R_{C-0}, R_{C-1}, \dots, R_{C-Nc-3}$ and R_{C-Nc-2} , and resistors R_{C-a} and R_{C-b} . Among them, $R_{C-a}=R_{C-b}=R, R_{C-i}=2^i R$ ($i=0, 1, N_{c-3}, N_{c-2}$), and R is unit resistance. The N_c bits of the binary code $C_{Nc-1}C_{Nc-2} \dots C_2C_1C_0$ control their corresponding switches, respectively. When one bit of $C_{Nc-2} \dots C_2C_1C_0$ is "0", its corresponding switch of $K_0, K_1, \dots, K_{Nc-2}$ is turned off. When one bit of $C_{Nc-2} \dots C_2C_1C_0$ is "1", its corresponding switch of $K_0, K_1, \dots, K_{Nc-2}$ is turned on, and the resistor connected in parallel to the switch is short circuited. When the bit C_{Nc-1} is "0", switch K_{Nc-1} is con-

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nected to V_{REF-} . When the bit C_{Nc-1} is "1", switch K_{Nc-1} is connected to V_{REF+} . Node C_a is connected to the first terminal of resistor R_{C-a} , and the other terminal of resistor R_{C-a} is connected to node A_0 . Further, node A_0 is connected to each of the first terminal of switch K_0 , the first terminal of resistor R_{C-0} , and the first terminal of resistor R_{C-b} . K_0 is controlled by the bit C_0 of the code value of N_c bits. The second terminal of resistor R_{C-b} is connected to node C_b . The second terminal of resistor R_{C-0} is connected to node A_1 . Further, node A_1 is connected to each of the second terminal of switch K_0 , the first terminal of switch K_1 , and the first terminal of resistor R_{C-1} . K_1 is controlled by the bit C_1 of the code value of N_c bits. The second terminal of resistor R_{C-1} is connected to node A_2 . The subsequent connections follow the previous pattern of connection. Node A_{Nc-3} is connected to each of the second terminal of switch K_{Nc-4} , the first terminal of switch K_{Nc-3} , and the first terminal of resistor R_{C-Nc-3} . K_{Nc-3} is controlled by the bit C_{Nc-3} of the code value of N_c bits. The second terminal of resistor R_{C-Nc-3} is connected to node A_{Nc-2} . Further, node A_{Nc-2} is connected to each of the second terminal of switch K_{Nc-3} , the first terminal of switch K_{Nc-2} , and the first terminal of resistor R_{C-Nc-2} . K_{Nc-2} is controlled by the bit C_{Nc-2} of the code value of N_c bits. The second terminal of resistor R_{C-Nc-2} is connected to node A_{Nc-1} . Further, node A_{Nc-1} is connected to each of the second terminal of the switch K_{Nc-2} and the first terminal of switch K_{Nc-1} . The second terminal and the third terminal of switch K_{Nc-1} are connected to V_{REF-} and V_{REF+} , respectively. Among them, the binary code of N_c bits is the second code value, and the N_c-1 is k .

Referring to FIG. 9, the third switch element **2241** may include a third inverter **207**, a third PMOS transistor **208**, and a third NMOS transistor **209**. The input of the third inverter **207** is used to receive the second code value, and the output of the third inverter **207** is connected to the gate of the third PMOS transistor **208**. The source of the third PMOS transistor **208** is connected to the drain of the third NMOS transistor **209**, and the drain of the third PMOS transistor **208** is connected to the source of the third NMOS transistor **209**.

As can be seen from FIG. 9, the input of the inverter **207** is connected to each of c_i ($i=0, 1, \dots, N_{c-2}$) in the N_c code value and the gate of the NMOS transistor **209** simultaneously. The output of the inverter **207** is connected to the gate of the PMOS transistor **208**. The source of the PMOS transistor **208** is connected to the drain of the NMOS transistor **209** and connected to node A_i ($i=0, 1, \dots, N_{c-1}$) in FIG. 8. The drain of the PMOS transistor **208** is connected to the source of the NMOS transistor **209** and connected to the node A_{i-1} ($i=0, 1, \dots, N_{c-1}$) in FIG. 8.

In the embodiment of the present disclosure, the output V_{dac-b} of the second adjustment unit **230** may be increased or decreased by changing the third code value, and the change is processed by the first adjustment unit **220**, to make V_{dac-a} perform the same increase or decrease based on V_{dac-b} . By changing the second code value, in the embodiment of the present disclosure, the minimum variation of V_{dac-a} and the minimum variation of V_{dac-b} may be changed simultaneously. In this way, the final output may be obtained, i.e., $V_{dac-a} + V_{dac-b} - V_{dac}$.

In the digital-to-analog conversion circuit provided by the embodiments of the present disclosure, the conversion unit and the first adjustment unit are incorporated to simply and effectively adjust a voltage range. Specifically, the digital-to-analog conversion circuit may include the conversion unit and the first adjustment unit. The conversion unit may include the first code value receiving terminal used to

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receive the first code value, the conversion unit is used to convert the first code value into an analog signal. The first adjustment unit may include the second code value receiving terminal used to receive the second code value. The signal input terminal of the first adjustment unit is connected to the signal output terminal of the conversion unit. The first adjustment unit is used to obtain the first analog adjustment signal based on the second code value, and adjust, with the first analog signal, the analog signal transmitted from the conversion unit, to obtain the target signal. In the present disclosure, after the conversion unit converts the received first code value into the analog signal, the analog signal may be adjusted by the first adjustment unit, which can make the finally obtained target signal accurate and effective. In addition, in the embodiments of the present disclosure, the conversion unit may serve as the main circuit and the second adjustment unit serve as the slave circuit, or the second adjustment unit may serve as the main circuit and the first adjustment unit may serve as the slave circuit; as such, the configuration of the digital-to-analog conversion circuit is flexible. Further, since the voltage range is enabled to being adjusted, the present disclosure may be applied to a real-time voltage calibration system.

Referring to FIG. 10, a method flowchart of a digital-to-analog conversion method according to an embodiment of the present disclosure is illustrated. The digital-to-analog conversion method may include blocks S210 and S220.

At block S210, a first code value is received, and converted into an analog signal.

In the embodiments of the present disclosure, when the first adjustment unit and the second adjustment unit are absent, the output of the conversion unit may be represented by the following formula:

$$V_{dac} = \sum_{k=1}^{N_a} \frac{a_{Na-k} \cdot V_{REF+} + a_{Na-k} \cdot V_{REF-}}{2^k} + \frac{V_{REF-}}{2^{N_a}}.$$

To better understand the correspondence between digital code values and analog code values, a diagram as illustrated in FIG. 11 is provided in the embodiment of the present disclosure. Due to the digital code, the output analog voltage cannot reach V_{REF+} . For the first code value (the code value of N_a bits), the number of the output analog voltages is $2^{N_a}-1$. The symbol "A" illustrated in FIG. 11 may be represented as

$$\frac{V_{REF+} - V_{REF-}}{2^{N_a}} + V_{REF-},$$

and the symbol "B" may be represented as

$$2 \cdot \frac{V_{REF+} - V_{REF-}}{2^{N_a}} + V_{REF-},$$

the symbol "C" may be represented as

$$(2^{N_a} - 2) \cdot \frac{V_{REF+} - V_{REF-}}{2^{N_a}} + V_{REF-},$$

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and the symbol "D" represents

$$(2^{N_a} - 1) \cdot \frac{V_{REF+} - V_{REF-}}{2^{N_a}} + V_{REF-}.$$

Further, "Y" illustrated in FIG. 11 represents

$$\frac{V_{REF+} - V_{REF-}}{2^{N_a}}.$$

In addition, when the second adjustment unit and the first adjustment unit are incorporated, the target signal (the total output V_{dac}) may be represented by the following formula:

$$V_{dac} = \frac{R}{R_{Ctotal} + R} \cdot (C_{Nc-1} \cdot V_{REF+} + \bar{C}_{Nc-1} \cdot V_{REF-}) + \frac{R_{Ctotal}}{R_{Ctotal} + R} \cdot \sum_{k=1}^{N_a} \frac{a_{Na-k} \cdot V_{REF+} + \bar{a}_{Na-k} \cdot V_{REF-}}{2^{k+1}} + \frac{R_{Ctotal}}{R_{Ctotal} + R} \cdot \frac{V_{REF-}}{2^{N_a+1}} + \frac{R_{Ctotal}}{R_{Ctotal} + R} \cdot \sum_{k=1}^{N_b} \frac{a_{Nb-k} \cdot V_{REF+} + \bar{a}_{Nb-k} \cdot V_{REF-}}{2^{k+1}} + \frac{R_{Ctotal}}{R_{Ctotal} + R} \cdot \frac{V_{REF-}}{2^{N_b+1}}.$$

In the above formula, the first term represents two ranges, i.e., a high range and a low range, controlled by the highest bit of the second code value, where the first term is

$$\frac{R}{R_{Ctotal} + R} \cdot (C_{Nc-1} \cdot V_{REF+} + \bar{C}_{Nc-1} \cdot V_{REF-});$$

and the second term and the third term represent the analog output voltage of the conversion unit, where the second term is

$$\frac{R_{Ctotal}}{R_{Ctotal} + R} \cdot \sum_{k=1}^{N_a} \frac{a_{Na-k} \cdot V_{REF+} + \bar{a}_{Na-k} \cdot V_{REF-}}{2^{k+1}},$$

and the third term is

$$\frac{R_{Ctotal}}{R_{Ctotal} + R} \cdot \frac{V_{REF-}}{2^{N_a+1}}.$$

In addition, the fourth term and the fifth term represent the output voltage of the second adjustment unit, where the fourth term is

$$\frac{R_{Ctotal}}{R_{Ctotal} + R} \cdot \sum_{k=1}^{N_b} \frac{a_{Nb-k} \cdot V_{REF+} + \bar{a}_{Nb-k} \cdot V_{REF-}}{2^{k+1}},$$

and the fifth term is

$$\frac{R_{Ctotal}}{R_{Ctotal} + R} \cdot \frac{V_{REF-}}{2^{N_b+1}}.$$

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Corresponding, the variation of the conversion unit is

$$1LSB = \frac{1}{2^{Na+1}} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \cdot (V_{REF+} - V_{REF-}),$$

and the minimum variation of the second adjustment unit is

$$1LSB = \frac{1}{2^{Nb+1}} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \cdot (V_{REF+} - V_{REF-}).$$

In addition, R_{Ctotal} is the total resistance of the N_c -bit 15 summation circuit, which is represented as

$$R_{Ctotal} = R_{Ctotal} = \sum_{k=1}^{N_c} \frac{C^{N_c-k}}{2^k} \cdot R. \quad 20$$

In some embodiments, when the highest bit C_{N_c-1} of the N_c bits is controlled to be “0”, the correspondence between the digital codes and the analog voltages is illustrated in FIG. 12. The second adjustment unit is controlled to adjust the output common mode voltage of the conversion unit, where the magnitude of adjusting the common mode voltage and the range of the common mode voltage may be controlled by the third code value. The symbol “A” illustrated in FIG. 12 may represent

$$\sum_{k=1}^{Nb} 2^{Nb-k} \cdot b_{Nb-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Nb+1}} + V_{REF-}; \quad 25$$

the symbol “B” may represent

$$\sum_{k=1}^{Nb} 2^{Nb-k} \cdot b_{Nb-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Nb+1}} + \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Na+1}} + V_{REF-}; \quad 30$$

the symbol “C” may represent

$$\sum_{k=1}^{Nb} 2^{Nb-k} \cdot b_{Nb-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Nb+1}} + \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Na+1}} + V_{REF-}; \quad 35$$

the symbol “D” may represent

$$\sum_{k=1}^{Nb} 2^{Nb-k} \cdot b_{Nb-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Nb+1}} + \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Na+1}} + (2^{Na} - 2) \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Na+1}} + V_{REF-}; \quad 40$$

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and the symbol “E” may represent

$$\sum_{k=1}^{Nb} 2^{Nb-k} \cdot b_{Nb-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Nb+1}} + \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Na+1}} + (2^{Na} - 1) \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Na+1}} + V_{REF-}. \quad 45$$

10 Further, the symbol “Y1” illustrated in FIG. 12 may represent

$$\sum_{k=1}^{Nb} 2^{Nb-k} \cdot b_{Nb-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Nb+1}},$$

and the symbol “Y2” illustrated in FIG. 12 may represent

$$\frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Na+1}}.$$

In some other embodiments, when the highest bit C_{N_c-1} of the second code value (the code value of N_c bits) is controlled to be “1”, the correspondence between the first code values (digital codes of N_a bits) and the analog voltages is illustrated in FIG. 13. In this case, the corresponding initial voltage is determined by the resistors controlled by the second code value (the code value of N_c bits). Similarly, the output common mode voltage of the conversion unit may be adjusted by controlling the third code value (the code value of N_b bits). The symbol “A” illustrated in FIG. 13 may represent

$$\sum_{k=1}^{Nb} 2^{Nb-k} \cdot b_{Nb-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Nb+1}} + \quad 45$$

$$\frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF+} + \frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF-}; \quad 50$$

the symbol “B” may represent

$$\sum_{k=1}^{Nb} 2^{Nb-k} \cdot b_{Nb-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Nb+1}} + \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Na+1}} + \frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF+} + \frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF-}; \quad 55$$

the symbol “C” may represent

$$\sum_{k=1}^{Nb} 2^{Nb-k} \cdot b_{Nb-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Nb+1}} + 2 \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Na+1}} + \frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF+} + \frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF-}; \quad 60$$

the symbol “D” may represent

$$\sum_{k=1}^{Nb} 2^{Nb-k} \cdot b_{Nb-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{Nb+1}} + \quad 65$$

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$$\begin{aligned} & \text{-continued} \\ & (2^{N_a} - 2) \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{N_a+1}} + \\ & \frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF+} + \frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF-}, \end{aligned} \quad 5$$

and the symbol “E” may represent

$$\begin{aligned} & \sum_{k=1}^{N_b} 2^{N_b-k} \cdot b_{N_b-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{N_b+1}} + \\ & (2^{N_a} - 1) \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{N_a+1}} + \\ & \frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF+} + \frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF-}. \end{aligned}$$

Further, the symbol “Y1” illustrated in FIG. 13 may represent

$$\frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF+} + \frac{R_{Ctotal}}{R_{Ctotal} + R} V_{REF-},$$

the symbol “Y2” illustrated in FIG. 13 may represent

$$\sum_{k=1}^{N_b} 2^{N_b-k} \cdot b_{N_b-k} \cdot \frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{N_b+1}},$$

and the symbol “Y3” illustrated in FIG. 13 may represent

$$\frac{R_{Ctotal}}{R_{Ctotal} + R} \frac{V_{REF+} - V_{REF-}}{2^{N_a+1}}.$$

At block S220, the analog signal is adjusted to obtain the target signal.

In some embodiments, the adjusting the analog signal to obtain the target signal may include: acquiring a digital-to-analog adjustment list, and obtaining, based on the digital-to-analog adjustment list, the target signal corresponding to the analog signal. The digital-to-analog adjustment list is obtained by acquiring multiple full-code-value digital signals, and adjusting analog signals corresponding to the multiple full-code-value digital signals.

As can be seen from the above introduction, in the embodiments of the present disclosure, in obtaining the target signal corresponding to the digital signal, a digital-to-analog adjustment list may be first created, and then the target signal corresponding to the analog signal is searched based on the digital-to-analog adjustment list. The digital-to-analog adjustment list may be obtained by acquiring multiple full-code-value digital signals and adjusting the analog signals corresponding to the multiple full-code-value digital signals. For example, as for the first code value, when $N_a=000$, $V_{OUT}=0V$; when $N_a=001$, $V_{OUT}=0.1V$; when $N_a=010$, $V_{OUT}=0.2V$; and when $N_a=011$, $V_{OUT}=0.3V$; through adjustment of the third code value N_b , when $N_a=000$, $V_{OUT}=0.2V$; when $N_a=001$, $V_{OUT}=0.3V$; when $N_a=010$, $V_{OUT}=0.4V$; and when $N_a=011$, $V_{OUT}=0.5V$; on this basis, through adjustment of the second code value N_c , when $N_a=000$, $V_{OUT}=0.2V$; when $N_a=001$, $V_{OUT}=0.25V$;

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when $N_a=010$, $V_{OUT}=0.3V$; and when $N_a=011$, $V_{OUT}=0.35V$. These data may be used to define the digital-to-analog adjustment list.

In the digital-to-analog conversion circuit and method provided by the embodiments of the disclosure, the conversion unit and the first adjustment unit are incorporated in the digital-to-analog conversion circuit to simply and effectively adjust the voltage range. Specifically, the digital-to-analog conversion circuit may include the conversion unit and the first adjustment unit. The conversion unit may include the first code value receiving terminal used to receive the first code value, and the conversion unit is used to convert the first code value into an analog signal. The first adjustment unit may include the second code value receiving terminal used to receive the second code value, and the signal input terminal of the first adjustment unit is connected to the signal output terminal of the conversion unit. The first adjustment unit is used to obtain a first analog adjustment signal based on the second code value, and adjust, with the first analog adjustment signal, the analog signal transmitted from the conversion unit, to obtain the target signal. In the present disclosure, after the conversion unit converts the received first code value into the analog signal, the analog signal may be adjusted by the first adjustment unit, which can make the finally obtained target signal accurate and effective.

As can be seen, it is understandable for those skilled in the art that, all or some steps in the disclosed method, the system disclosed above, and functional modules/units in the system may be implemented in software (which may be implemented by computer program codes executable by the computing system), firmware, hardware, or an appropriate combination thereof. In a hardware implementation, the division between the functional modules/units mentioned in the above description may not necessarily correspond to the division of physical components; for example, one physical component may have multiple functions, or one function or step may be cooperatively executed by several physical components. Some or all physical components may be implemented as software executed by a processor, such as a central processor, a digital signal processor, or a microprocessor, or be implemented as hardware, or be implemented as an integrated circuit such as an application-specific integrated circuit.

In addition, as known to those skilled in the art, a communication medium typically includes computer-readable instructions, data structures, computer program modules, or other data in which data signals are modulated such as a carrier or other transmission mechanisms, and may include any information transferring medium. Therefore, the present disclosure is not limited to any combination of specific hardware and software.

The foregoing is detailed descriptions, made in conjunction with the specific implementations, of the embodiments of the present disclosure, and the specific embodiments of the present disclosure cannot be considered as being limited to these descriptions. For those of ordinary skill in the art to which the present disclosure is related, some simple deductions or substitutions may be made without departing from the concept of the present disclosure, and they should also be considered as falling within the scope of protection of the present disclosure.

What is claimed is:

1. A digital-to-analog conversion circuit, comprising:
 - a conversion unit comprising a first code value receiving terminal, wherein the first code value receiving terminal is configured to receive a first code value, and the

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conversion unit is configured to convert the first code value into an analog signal; and
 a first adjustment unit comprising a second code value receiving terminal, wherein the second code value receiving terminal is used to receive a second code value, a first signal input terminal of the first adjustment unit is connected to a signal output terminal of the conversion unit, and the first adjustment unit is configured to: obtain a first analog adjustment signal according to the second code value, and adjust, with the first analog adjustment signal, the analog signal transmitted from the conversion unit, to obtain a target signal;

wherein the first adjustment unit comprises k third resistor-selection sub-units, each of the third resistor-selection sub-units comprises a third switch element and a fifth resistor, and the third switch element is connected in parallel to the fifth resistor.

2. The digital-to-analog conversion circuit of claim 1, wherein the digital-to-analog conversion circuit further comprises a second adjustment unit, the second adjustment unit comprises a third code value receiving terminal configured to receive a third code value; a signal output terminal of the second adjustment unit is connected to a second signal input terminal of the first adjustment unit; and the second adjustment unit is configured to obtain a second analog signal according to the third code value, and adjust, with the second analog signal, the analog signal transmitted by the conversion unit.

3. The digital-to-analog conversion circuit of claim 2, wherein the second adjustment unit comprises n first resistor-selection sub-units, a first resistor is provided between every two adjacent first resistor-selection sub-units, and the first resistor is electrically connected to its adjacent first resistor-selection sub-units.

4. The digital-to-analog conversion circuit of claim 3, wherein each of the first resistor-selection sub-units comprises a second resistor and a first switch element, a control terminal of the first switch element is configured to receive the third code value, a second connection terminal of the first switch element is connected to a first connection terminal of the second resistor, and a second connection terminal of the second resistor is connected to the first resistor.

5. The digital-to-analog conversion circuit of claim 4, wherein the control terminal of each first switch element is configured to receive a bit of the third code value;

when the received bit is 0, the first switch element is adapted to be controlled to be turned off and connected to a negative pole terminal of a reference voltage; and
 when the received bit is 1, the first switch element is adapted to be controlled to be turned on and connected to a positive pole terminal of the reference voltage.

6. The digital-to-analog conversion circuit of claim 4, wherein the first switch element comprises a first inverter, a first PMOS transistor, and a first NMOS transistor, an input terminal of the first inverter is configured to receive the third code value, an output terminal of the first phase inverter is connected to each of a gate of the first PMOS transistor and a gate of the first NMOS transistor, a source of the first PMOS transistor is connected to a positive pole terminal of a reference voltage, a drain of the first PMOS transistor is connected to a drain of the first NMOS transistor, and a source of the first NMOS transistor is connected to a negative pole terminal of the reference voltage.

7. The digital-to-analog conversion circuit of claim 1, wherein the conversion unit comprises m second resistor-selection sub-units, a third resistor is provided between

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every two adjacent second resistor-selection sub-units, and the third resistor is electrically connected to its adjacent second resistor-selection sub-units.

8. The digital-to-analog conversion circuit of claim 7, wherein each of the second resistor-selection sub-units comprises a fourth resistor and a second switch element, a control terminal of the second switch element is configured to receive the first code value, a second connection terminal of the second switch element is connected to a first connection terminal of the fourth resistor, and a second connection terminal of the fourth resistor is connected to the third resistor.

9. The digital-to-analog conversion circuit of claim 8, wherein the control terminal of each second switch element is configured to receive a bit of the first code value;

when the received bit is 0, the second switch element is adapted to be controlled to be turned off and connected to a negative pole terminal of a reference voltage, and
 when the received bit is 1, the second switch element is adapted to be controlled to be turned on and connected to a positive pole terminal of the reference voltage.

10. The digital-to-analog conversion circuit of claim 8, wherein the second switch element comprises a second inverter, a second PMOS transistor, and a second NMOS transistor, an input terminal of the second inverter is configured to receive the first code value, an output terminal of the second inverter is connected to each of a gate of the second PMOS transistor and a gate of the second NMOS transistor, a source of the second PMOS transistor is connected to a positive pole terminal of a reference voltage, a drain of the second PMOS transistor is connected to a drain of the second NMOS transistor, and a source of the second NMOS transistor is connected to a negative pole terminal of the reference voltage.

11. The digital-to-analog conversion circuit of claim 1, wherein a control terminal of each third switch element is configured to receive a bit of the second code value;

when the received bit is 0, the third switch element is adapted to be controlled to be turned off; and

when the received bit is 1, the third switch element is adapted to be controlled to be turned on, and the fifth resistor connected in parallel to the third switch element is short circuited.

12. The digital-to-analog conversion circuit of claim 1, wherein the third switch element comprises a third inverter, a third PMOS transistor, and a third NMOS transistor, an input terminal of the third inverter is configured to receive the second code value, an output terminal of the third inverter is connected to a gate of the third PMOS transistor, a source of the third PMOS transistor is connected to a drain of the third NMOS transistor, and a drain of the third PMOS transistor is connected to a source of the third NMOS transistor.

13. A digital-to-analog conversion method, implemented by a digital-to-analog conversion circuit comprising a conversion unit and a first adjustment unit, the method comprising:

receiving, by the conversion unit, a first code value, and
 converting the first code value into an analog signal;

receiving, by the first adjustment unit, a second code value;

obtaining, by the first adjustment unit, a first analog adjustment signal according to the second code value; and

adjusting, by the first adjustment unit with the first analog adjustment signal, the analog signal obtained from the conversion unit, to obtain a target signal;

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wherein the first adjustment unit comprises k third resistor-selection sub-units, each of the third resistor-selection sub-units comprises a third switch element and a fifth resistor connected in parallel to each other, a control terminal of each third switch element is configured to receive a bit of the second code value, and the method further comprises:

in response to detecting that the received bit is 0, controlling the third switch element to be turned off; and

in response to detecting that the received bit is 1, controlling the third switch element to be turned on, to make the fifth resistor connected in parallel to the third switch element be short circuited.

14. The digital-to-analog conversion method of claim **13**, wherein the digital-to-analog conversion circuit further comprises a second adjustment unit, and the method further comprises:

receiving, by the second adjustment signal, a third code value;

obtaining a second analog signal according to the third code value; and

adjusting, with the second analog signal, the analog signal obtained by the conversion unit.

15. The digital-to-analog conversion method of claim **14**, wherein the second adjustment unit comprises n first resistor-selection sub-units, a first resistor is electrically connected between every two adjacent first resistor-selection sub-units, each of the first resistor-selection sub-units comprises a second resistor and a first switch element, a control terminal of the first switch element is configured to receive a bit of the third code value, a second connection terminal of the first switch element is connected to a first connection

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terminal of the second resistor, a second connection terminal of the second resistor is connected to the first resistor, and the method further comprises:

in response to detecting that the received bit is 0, controlling the first switch element to be turned off and connected to a negative pole terminal of a reference voltage; and

in response to detecting that the received bit is 1, controlling the first switch element to be turned on and connected to a positive pole terminal of the reference voltage.

16. The digital-to-analog conversion method of claim **14**, wherein the conversion unit comprises m second resistor-selection sub-units, a third resistor is electrically connected between every two adjacent second resistor-selection sub-units, each of the second resistor-selection sub-units comprises a fourth resistor and a second switch element, a control terminal of the second switch element is configured to receive a bit of the first code value, a second connection terminal of the second switch element is connected to a first connection terminal of the fourth resistor, and a second connection terminal of the fourth resistor is connected to the third resistor, and the method further comprises:

in response to detecting that the received bit is 0, controlling the second switch element to be turned off and connected to a negative pole terminal of a reference voltage, and

in response to detecting that the received bit is 1, controlling the second switch element to be turned on and connected to a positive pole terminal of a reference voltage.

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